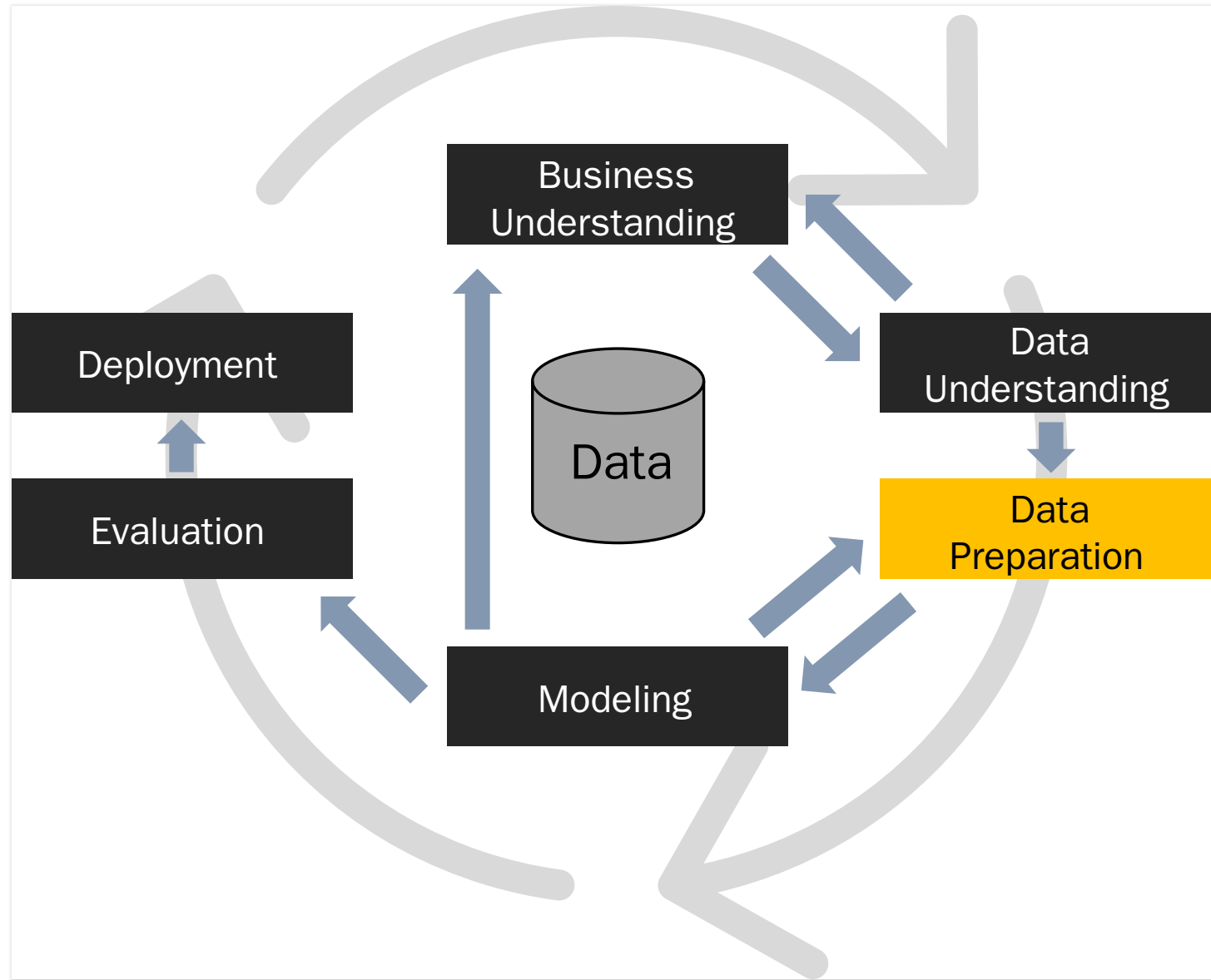


Machine Learning and Data Mining (Module 2)

Data Preparation

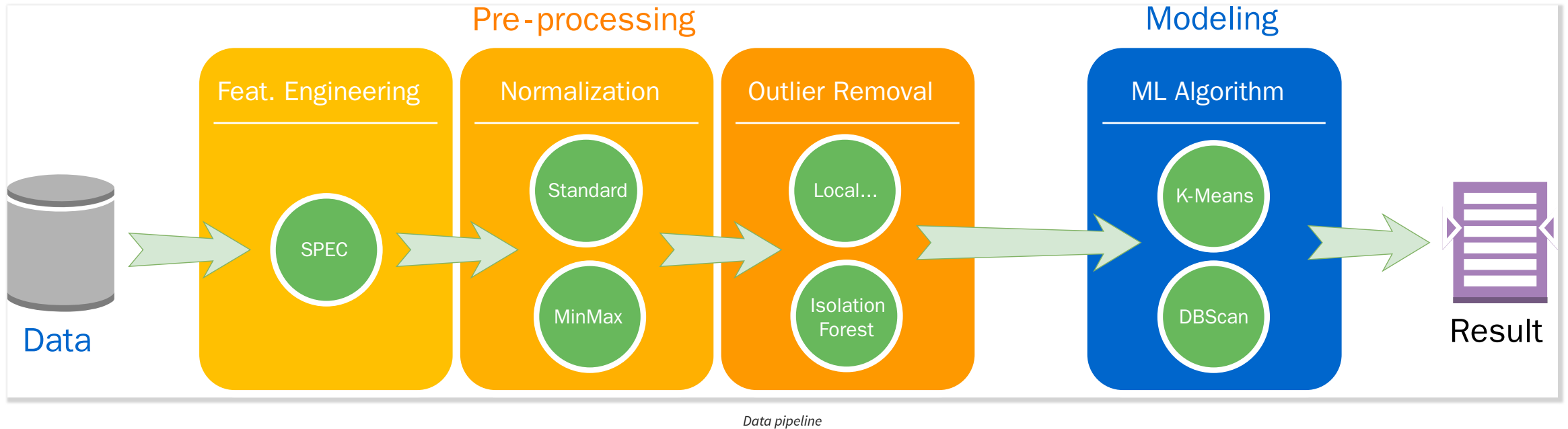
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Without clean data, the results of a data mining analysis are in question.

Data Preparation (aka *data pre-processing*)

The **data preparation** phase covers all activities to construct the dataset fed into the modeling tools from the initial data.



It plays a key role in a data analytics process and *avoids “garbage in, garbage out”*

- *A broad range of activities*; from correcting errors to selecting the most relevant features
 - Out-of-range values (`Income: -100`), Impossible combinations (`Exam mark: 15` and `Exam result: Passed`), Missing values...
- Data scientists *cannot easily foresee the impact of pipeline prototypes*
 - There are *no pre-defined rules* on the impact of pre-processing transformations
 - Outline how each quality problem (reported in the earlier “Verify Data Quality” step) has been addressed

Data pre-processing includes ([Shearer 2000](#)) data

1. *selection*
2. *cleansing*
3. *construction*
4. *integration*
5. *formatting*



Problem: which data should we use?

Select Data

Deciding on the data that will be used for the analysis is based on several criteria, including

- its *relevance* to the data mining goals
- *quality* and *technical* constraints such as GDPR or limits on data volume or data types

Part of the data selection process should involve explaining why certain data was included or excluded.

- It is also a good idea to decide if one or more attributes are more important than others

Examples

- If decisions are based on the geographical region, individuals' addresses can be dropped to reduce the amount of data.

<u>Address</u>	<u>Country</u>	<u>Sales</u>
Via dell'Università 50	Italy	1000
...	...	...

- To learn how sales are characterized by store *Type*, you do not need to consider the *StoreId*

<u>StoreId</u>	<u>Type</u>	<u>Sales</u>
S1	grocery	1000
S2	supermarket	1500
S3	...	...

Select Data

[Flight delays dataset](#) contains information about flights in the United States.

- Task predict flight delays ([ARR_DELAY](#): Arrival delay) based on the following (31) features
- Features
 1. [FL_DATE](#): Date of the flight.
 2. [AIRLINE](#): Name of the airline.
 3. [AIRLINE_DOT](#): DOT identifier for the airline.
 4. [AIRLINE_CODE](#): Code assigned to the airline.
 5. [DOT_CODE](#): DOT identifier.
 6. [FL_NUMBER](#): Flight number.
 7. [ORIGIN](#): Origin airport code.
 8. [ORIGIN_CITY](#): City of origin airport.
 9. [DEST](#): Destination airport code.
 10. [DEST_CITY](#): City of destination airport.
 11. [DEP_TIME](#): Actual departure time.
 12. [DEP_DELAY](#): Departure delay.
 13. [ARR_TIME](#): Actual arrival time.
 14. ...

What features would you select/drop to predict flight delays?

What personal data is considered sensitive?

The following personal data is considered *sensitive* and is subject to specific processing conditions:

- personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs;
- trade-union membership;
- genetic data, biometric data processed solely to identify a human being;
- health-related data;
- data concerning a person's sex life or sexual orientation.

See [article 4](#)

Why should we care about sensitive data?

The Artificial Intelligence Act

The *AI Act* is a European Union regulation concerning artificial intelligence that classifies applications by their risk of causing harm.

- **Unacceptable-risk** applications are *banned*.
 - Applications that manipulate human behaviour, use real-time remote biometric identification in public spaces, and social scoring (ranking individuals based on their personal characteristics, socio-economic status, or behaviour)
- **High-risk** applications must comply with *security*, *transparency* and *quality obligations*, and undergo conformity assessments.
 - AI applications that are expected to pose significant threats to health, safety, or the fundamental rights of persons.
 - They must be evaluated both before they are placed on the market and throughout their life cycle.
- **Limited-risk** applications *only have transparency* obligations.
 - AI applications that make it possible to generate or manipulate images, sound, or videos.
 - Ensure that users are informed that they are interacting with an AI system and are allowed to make informed choices.
- **Minimal-risk** applications *are not regulated*.
 - AI systems used for video games or spam filters.



AI Act and Black Mirror

See *Black Mirror* episodes and how they relate to the AI Act's high-risk categories:

- “The Entire History of You” (Season 1, Episode 3)
 - People have **memory implants** that allow them to replay and analyze past events.
- “Nosedive” (Season 3, Episode 1)
 - People’s **social credit scores** determine their access to housing, jobs, and even flights.
- “Hated in the Nation” (Season 3, Episode 6)
 - A social media campaign with an AI-driven hashtag leads to **automated drone assassinations**.
- “Metalhead” (Season 4, Episode 5)
 - The episode features relentless **autonomous killer robots** that hunt down humans.
- “Rachel, Jack and Ashley Too” (Season 5, Episode 3) – AI & Digital Manipulation
 - A pop star’s **consciousness is cloned into an AI assistant**, and the AI is used to create performances without her consent.



Problem: missing values?

A retail company tracks daily sales, but some records are missing due to system failures.

A telecom company is predicting customer churn, but some customers have missing contract durations or monthly bill values.

A hospital maintains records of patients' blood pressure, but 15% of entries are missing.

A bank evaluates loan applications, but some applicants have missing income data.



Problem: missing values?

A retail company tracks daily sales, but some records are missing due to system failures.

- *Use historical sales trends* to impute missing values

A telecom company is predicting customer churn, but some customers have missing contract durations or monthly bill values.

- *Use median imputation for numerical features* (e.g., replace missing monthly bill amounts with the median).

A hospital maintains records of patients' blood pressure, but 15% of entries are missing.

- Use K-Nearest Neighbors (KNN) imputation to *estimate missing values based on similar patients*.

A bank evaluates loan applications, but some applicants have missing income data.

- *Use group-based imputation* (e.g., average income for self-employed individuals).

Imputation of missing values

Imputation is the process of replacing missing data with substituted values.

Listwise deletion (complete case) deletes data with missing values

- If data are missing at random, listwise deletion does not add any bias, but it decreases the sample size
- Otherwise, listwise deletion will introduce bias because the remaining data are not representative of the original sample

Before cleaning

StoreId	type	sales
S1		1000
S2	supermarket	
S3	grocery	100

After cleaning

StoreId	type	sales
S3	grocery	100

Pairwise deletion deletes data when it is missing a variable required for a particular analysis

- ... but includes that data in analyses for which all required variables are present

Imputation of missing values

Hot-deck imputation: the information donors come from the same dataset as the recipients

- One form of hot-deck imputation is called “last observation carried forward”
 - Sort a dataset according to any number of variables, thus creating an ordered dataset
 - Finds a missing value and uses the value immediately before the data that is missing to impute the missing value

Before cleaning

StoreId	Date	sales
S1	2024-10-04	1000
S1	2024-10-05	
S2	2024-01-04	

After cleaning (sort by `StoreId` and `Date`)

StoreId	Date	sales
S1	2024-10-04	1000
S1	2024-10-05	1000
S2	2024-01-04	1000

Cold-deck imputation replaces missing values with values from similar data in different datasets

Imputation of missing values

Mean substitution replaces missing values with the mean of that variable for all other cases

- Mean imputation attenuates any correlations involving the variable(s) that are imputed
 - There is no relationship between the imputed variable and any other measured variables.
 - Mean imputation can be carried out within classes (i.e., categories, such as gender)

Before cleaning

StoreId	Date	sales
S1	2024-10-04	1000
S1	2024-10-05	
S1	2024-10-06	2000
S2	2024-10-04	
S2	2024-10-05	1000

After cleaning (average by `StoreId`)

StoreId	Date	sales
S1	2024-10-04	1000
S1	2024-10-05	1500
S1	2024-10-06	2000
S2	2024-10-04	1000
S2	2024-10-05	1000

Case Study: How Do We Impute?

Year	Portfolio Value
2008	1000.00
2009	1050.00
2010	1102.50
2011	1157.63
2012	
2013	1276.28
2014	1340.10
2015	1407.10
2016	1477.46
2017	1551.33
2018	
2019	1710.34
2020	1795.86
2021	1885.65
2022	1979.93
2023	
2024	2182.87

Case Study: Compound Interest

Our portfolio has an initial value $V_0 = 1000\text{€}$ and each has a return of $X\%$

The first year, the portfolio increases its value to $V_1 = 1000\text{€} + (1000\text{€} \times X\%) = 1050\text{€}$

The second year, the portfolio increases its value to $V_2 = V_1 + (V_1 \times X\%) = 1102.50\text{€}$

... and so on.

Year	Value
0	1000.00 €
1	1050.00 €
2	1102.50 €
...	...
18	2406.62 €

This is not a linear increase but a geometric sequence: $\text{Final value} = \text{Initial value} \times \left(1 + \frac{r}{n}\right)^{\frac{t}{n}}$

- r is the nominal annual interest rate
- n is the compounding frequency (1: annually, 12: monthly, 52: weekly, 365: daily)
- t is the overall length of time the interest is applied (expressed using the same time units as n , usually years).

$$\text{Final value} = \text{Initial value} \times (1 + X)^{18}$$

$$X = \left(\frac{\text{Final value}}{\text{Initial value}}\right)^{\frac{1}{18}} - 1$$

$$X = 0.05 = 5\%$$

In this case, the return is equal every year, so the average interest is 5%.

Case Study: Compound Interest

Let us assume now that the returns change over the year.

Year	Value	Return
0	1000.00 €	-
1	1050.00 €	5%
2	997.50 €	-5%
3	1047.38 €	5%
4	995.01 €	-5%
5	995.01 €	0%

What is the average return?

Case Study: Compound Interest

If we apply the arithmetic mean it is $\frac{5-5+5-5+0}{5} = 0$

However, **this is wrong!**

We already know that $X = \frac{\text{Final value}}{\text{Initial value}}^{\frac{1}{5}} - 1$

The average return is $X = -0.1\%$

$X\%$ is the *Compound Annual Growth Rate*, the mean annualized growth rate for compounding values over a given time period.

- CAGR smoothes the effect of the volatility of periodic values that can render arithmetic means less meaningful.

Year	Value	Return
0	1000.00 €	-
1	1000.00 €	0%
2	1000.00 €	0%
...	...	...
5	1000.00 €	0%

Year	Value	Return
0	1000.00 €	-
1	999.00 €	-0.1%
2	998.00 €	-0.1%
...	...	...
5	995.01 €	-0.1%

End of the Case Study

Takeaway: pay attention to the semantics of the features!



Problem: how do we handle anomalies?

A customer who typically buys groceries worth \$50 suddenly places an order for \$5,000 in electronics. Is it a fraud?

A system that usually receives 100-200 requests per second suddenly sees 10,000 requests per second. Is it a DDoS attack?

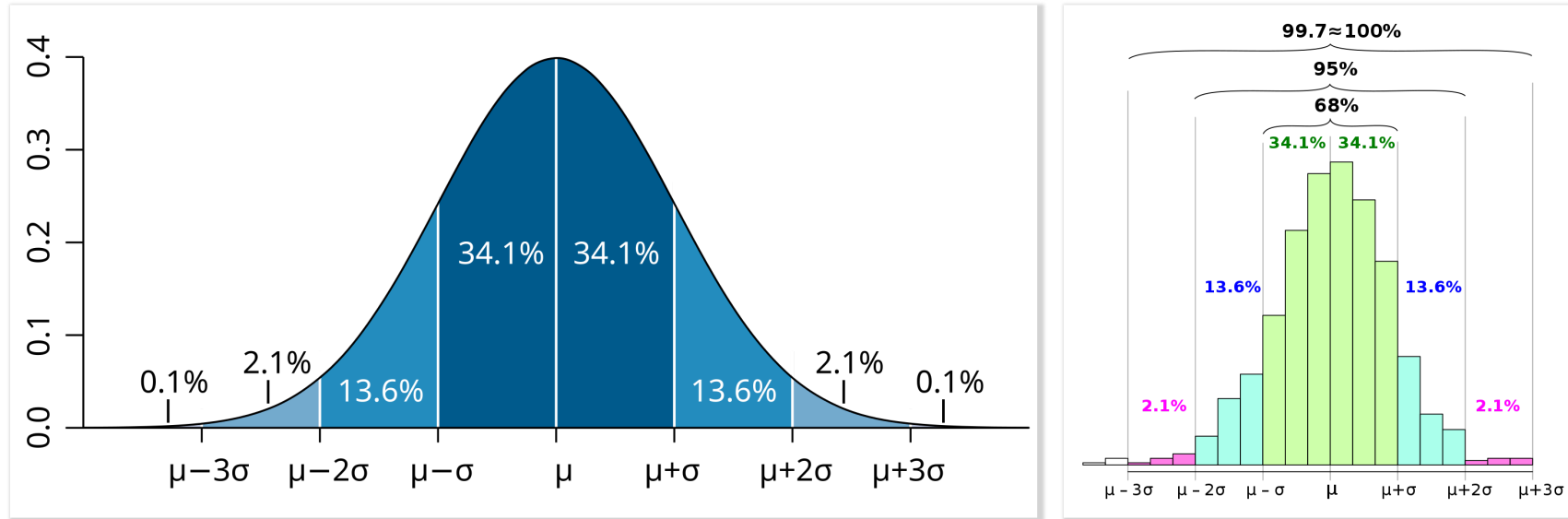
A bottle-filling machine fills 500ml of liquid, but occasionally, some bottles contain 450ml or 550ml. Is it a defect?

A credit card transaction of \$10,000 when the usual spending is around \$50-\$100. Is it a fraud?

Outlier removal

Outlier removal is the process of eliminating data points that deviate significantly from the rest of the dataset

- An *outlier* is a data point that differs significantly from other observations (e.g., a measurement error or heavy-tailed distribution)



In the case of normally distributed data, the *three sigma rule* means that:

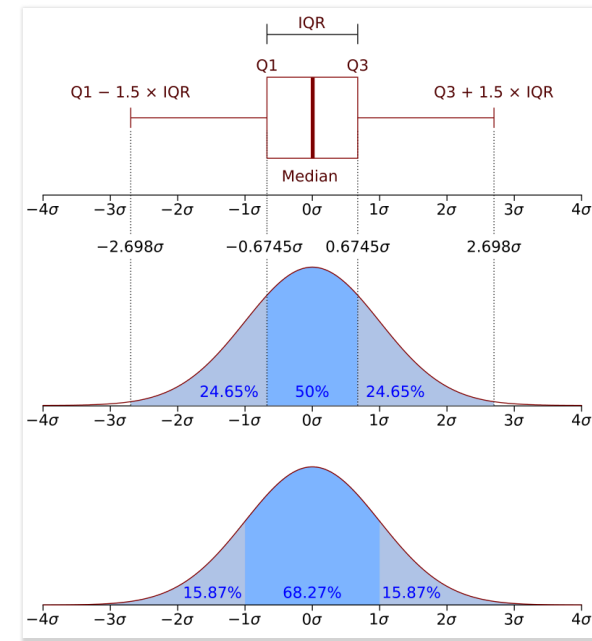
- Nearly all values (99.7%) lie within three standard deviations of the mean
 - Roughly 1 in 22 observations will differ by twice the standard deviation or more from the mean,
 - ... and 1 in 370 will deviate by three times the standard deviation.
- If the sample size is only 100, however, just three such outliers are already a reason for concern.

Outlier removal

Other methods flag outliers based on measures such as the interquartile range.

If Q_1 and Q_3 are the lower and upper quartiles, an outlier is any observation outside the range:

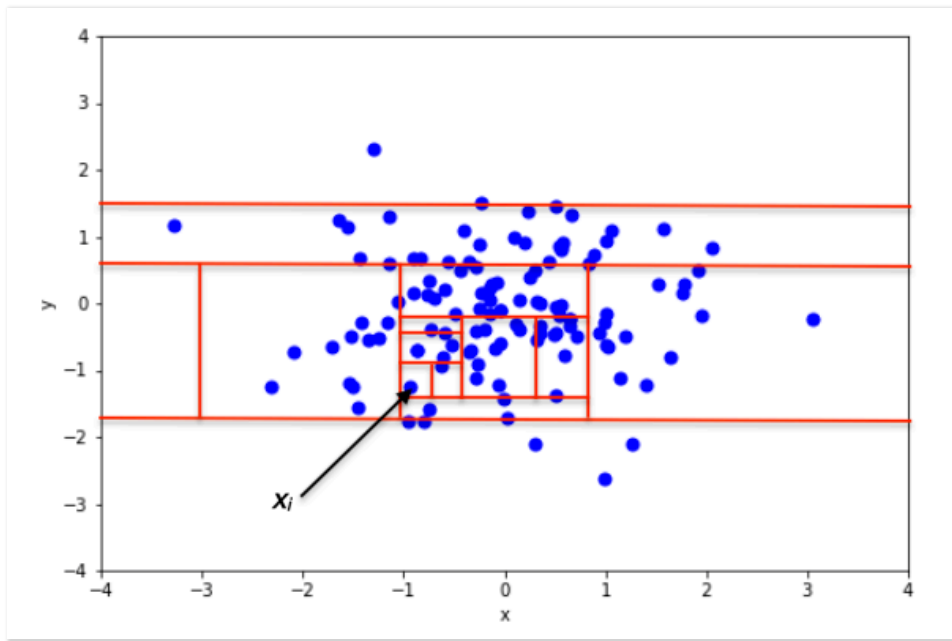
- $[Q_1 - k(Q_3 - Q_1), Q_3 + k(Q_3 - Q_1)]$ for some nonnegative k
- $k = 1.5$ to indicate an “outlier”
- $k = 3$ to indicate data that is “far out”



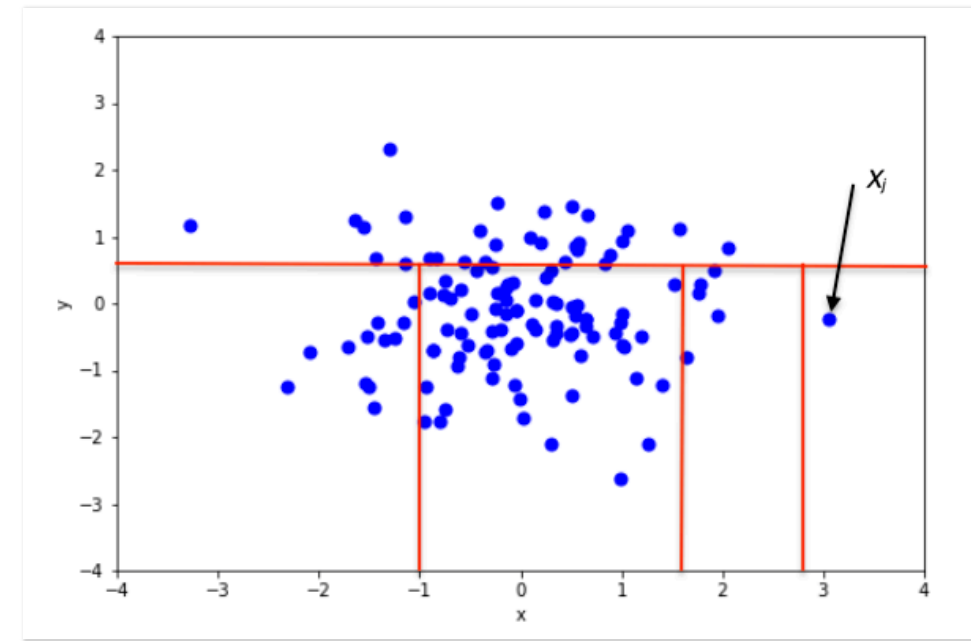
Outlier removal

Isolation Forest (Liu et al. 2008) is an algorithm for data anomaly detection using binary trees

- Because anomalies are few and different from other data, they can be isolated using a few partitions.
- Unlike decision tree algorithms, it uses only path length to output an anomaly score.



Inlier



Outlier

Case Study: The Black Swan Theory

Juvenal (55-128, Roman poet) wrote in his Satire VI of *events being* “a bird as rare upon the earth as a black swan”

- When the phrase was coined, the black swan was presumed by the Romans not to exist.
- All swans are white because all records reported that swans had white feathers.

In 1697, Dutch explorers became the first Europeans to see black swans in Australia.

- Observing a *single black swan* is the undoing of the logic of any system of thought
 - ... as well as any reasoning that followed from that underlying logic.
- Conclusions are potentially undone once any of its fundamental postulates is disproved.



Black swan

The black swan theory was developed by Nassim Nicholas Taleb ([Taleb 2008](#)) to explain:

- The disproportionate role of hard-to-predict and rare events that are beyond the realm of normal expectations.
- The non-computability of the probability of consequential rare events using scientific methods.
- The psychological biases that blind people to uncertainty and to the substantial role of rare events in historical affairs.

Such extreme events (outliers) collectively play vastly larger roles than regular occurrences.

Case Study: The Black Swan Theory

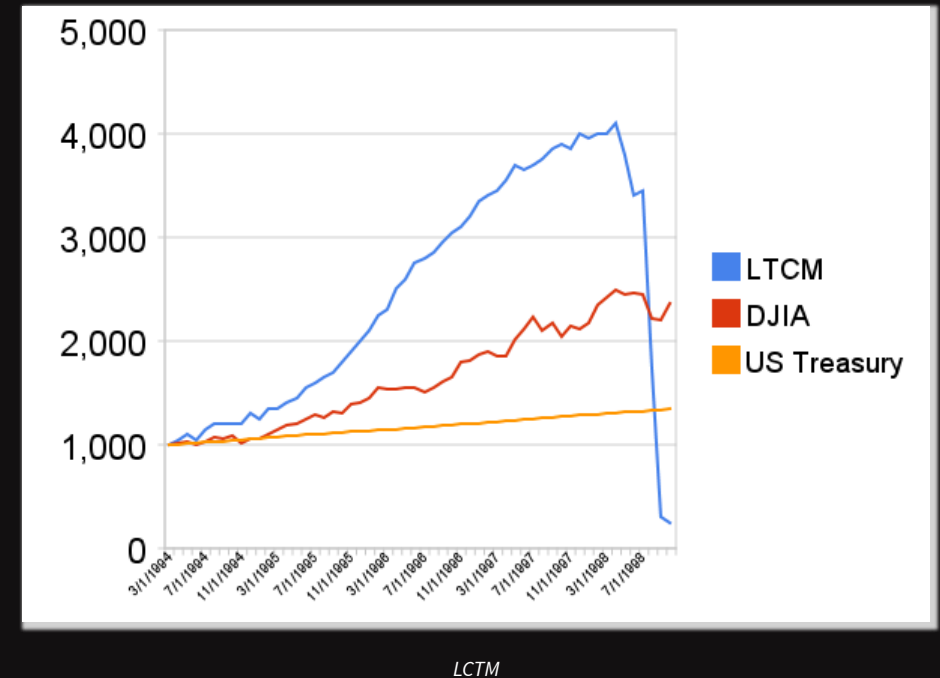
Long-Term Capital Management was a *highly leveraged* hedge fund.

- Members of LTCM's board of directors included Myron Scholes and Robert C. Merton, who shared the Nobel Prize in Economics
- LTCM was initially successful, with annualized returns of around 21% in its first year, 43% in its second year, and 41% in its third year.
- In 1998, **it lost \$4.6 billion** *in less than four months due to an unlikely combination* of (1997) Asian and (1998) Russian financial crises.

(Jorion 2000) [...] on 21 August, the portfolio lost \$550 million. By 31 August, the portfolio had lost \$1,710 million in 1 month.

- Using the presumed \$45 million daily (or \$206 million monthly) standard deviation, this translates into an *8.3 standard deviation event*.
- Assuming a normal distribution, *such an event would occur once every 800 trillion years, or 40,000 times the age of the universe*.

Surely this assumption was wrong.





Problem: is the dataset ready for machine learning?

An online retailer wants to predict which customers are likely to churn. Instead of using raw purchase data, they need a “Loyalty Score” based on Total purchases in the last 12 months, average order value, and frequency of purchases.

A bank needs to classify customers into risk levels based on their credit score.

Netflix needs to recommend movies based on genre. However, movie genres are categorical (e.g., “Action,” “Comedy”), which must be converted into numbers.

Feature engineering

Feature engineering refers to the manipulation (addition, deletion, combination, mutation) of your data set to improve machine learning model training.

Derived attributes should be added if they ease the modeling algorithm

```
Area = Length x Width.
```

```
Loyalty_Score = Total_Purchases x 0.4 + Avg_Order_Value x 0.3 + Frequency x 0.3.
```

```
Ocean_Proximity = distance((Ocean_Latitude, Ocean_Longitude), (District_Latitude, District_Longitude)).
```

Encoding may be necessary to transform *or symbolic fields* (“definitely yes”, “yes”, “don’t know”, “no”) to numeric values

Encoding

Encoding is the process of converting categorical variables into numeric features.

- Most machine learning algorithms, like linear regression and support vector machines, require input data to be numeric because they use numerical computations to learn the model.
- These algorithms are not inherently capable of interpreting categorical data.
- Some implementations of decision tree-based algorithms can directly handle categorical data.

Categorical features can be *nominal* or *ordinal*.

- *Nominal features* (e.g., colors) do not have a defined ranking or inherent order.
- *Ordinal features* (e.g., size) have an inherent order or ranking

One hot encoding and *ordinal encoding* are the most common methods to transform categorical variables into numerical features.

Encoding: ordinal encoding

Ordinal encoding replaces each category with an integer value.

- These numbers are, in general, assigned arbitrarily.
- Ordinal encoding is a preferred option when the categorical variable has an inherent order.

Before encoding

ProductId	Size
P1	small
P2	medium
P3	large
P4	small

After encoding (small = 0, medium = 1, large = 2)

ProductId	Size	Size_Enc
P1	small	0
P2	medium	1
P3	large	2
P4	small	0

Encoding: Likert scale

The Likert scale is widely used in social work research and is commonly constructed with four to seven points.

- [`*`, `**`, `***`, `****`, `*****`]
- [`1`, `2`, `3`, `4`, `5`]

What about averaging?

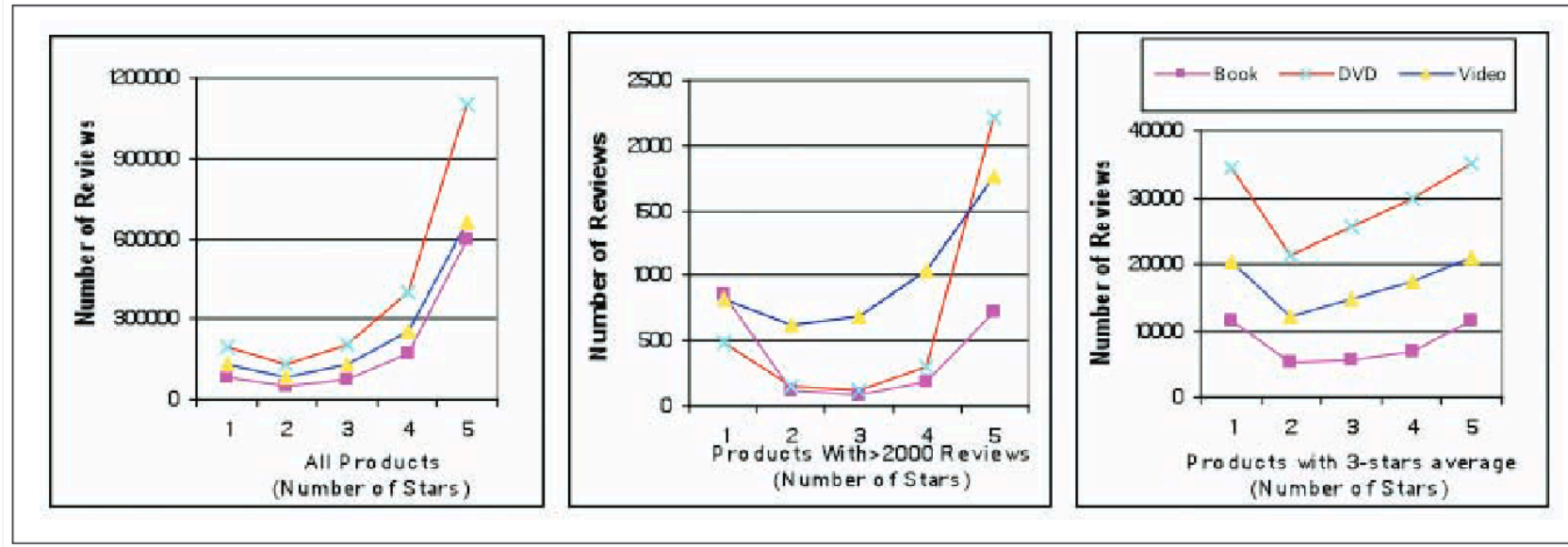
Encoding: Likert scale

It is usually treated as an interval scale, but strictly speaking, it is an ordinal scale, where arithmetic operations cannot be conducted ([Wu and Leung 2017](#))

Converting responses to a Likert-type question into an average seems an obvious and intuitive step, but it doesn't necessarily constitute good methodology. One important point is that respondents are often reluctant to express a strong opinion and may distort the results by gravitating to the neutral midpoint response. It also assumes that the emotional distance between mild agreement or disagreement and strong agreement or disagreement is the same, which isn't necessarily the case. At its most fundamental level, the problem is that the numbers in a Likert scale are not numbers as such, but a means of ranking responses.

Encoding: Likert scale

Figure 1. The Distribution of Product Reviews on Amazon



J-shaped distribution

People tend to write reviews only when they are either extremely satisfied or extremely unsatisfied.

People who feel the product is average might not bother to write a review.

([Hu et al. 2009](#))

Encoding: one-hot encoding

One-hot encoding (OHE) replaces categorical variables by *a set of binary variables* (each representing a category in the variable)

- The binary variable takes the value 1 if the observation shows the category, or alternatively, 0.
- One hot encoding treats each category independently.

Before encoding

ProductId	Color
P1	red
P2	green
P3	blue
P4	red

After encoding

ProductId	Color	red	green	blue
P1	red	1	0	0
P2	green	0	1	0
P3	blue	0	0	1
P4	red	1	0	0

OHE increases the dimensionality of the dataset, and it may not be suitable for encoding high cardinality features.

- To prevent a massive increase in the feature space, we can one-hot encode only the most frequent categories in the variable.
- ... less frequent values are treated collectively and represented as 0s in all the binary variables.

Case Study: Encoding Wrong Data Types

Microsoft has released an emergency fix for a year 2022 bug that is breaking email delivery on on-premise Microsoft Exchange servers.

As the year 2022 rolled in and the clock struck midnight, Exchange admins worldwide discovered that their servers were no longer delivering email. After investigating, they found that mail was getting stuck in the queue, and the Windows event log showed one of the following errors.

```
Log Name: Application
Source: FIPFS
Logged: 1/1/2022 1:03:42 AM
Event ID: 5300
Level: Error
Computer: server1.contoso.com
Description: The FIP-FS "Microsoft" Scan Engine failed to load. PID: 23092, Error
Code: 0x80004005. Error Description: Can't convert "2201010001" to long.
```

```
Log Name: Application
Source: FIPFS
Logged: 1/1/2022 11:47:16 AM
Event ID: 1106
Level: Error
Computer: server1.contoso.com
Description: The FIP-FS Scan Process failed initialization. Error: 0x80004005. Err
or Details: Unspecified error.
```

Y2K22 bug

Case Study: Encoding Wrong Data Types

Encoding the date 2022-01-01T00:01 into a signed integer 2201010001

A signed integer is a 32-bit datum that represents an integer in the range:

- Valid range: $[-2^{31}, 2^{31} - 1] = [-2147483648, 2147483647]$
- However, $2201010001 > 2147483647$

See also the [year 2000 problem](#)



Y2K



Problem: how do we treat features with different ranges?

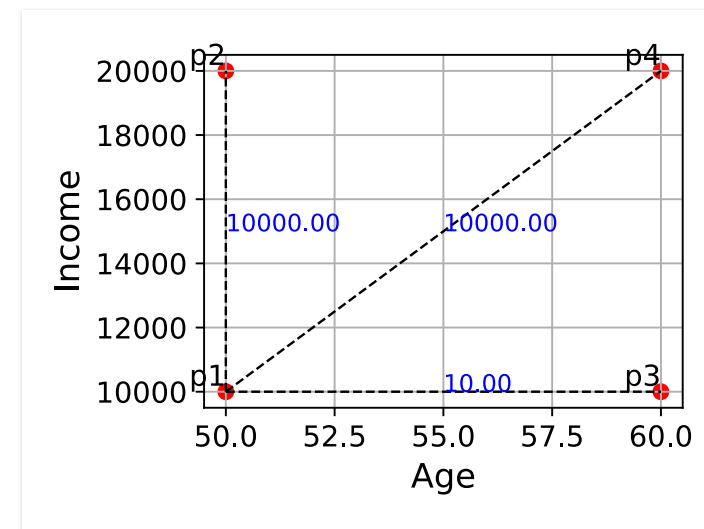
If data values vary widely, objective functions will not work properly without normalization in some ML algorithms.

- For example, many classifiers calculate the distance between two points using the Euclidean distance.
 - $d(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \dots + (p_n - q_n)^2} = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$
- If one of the features has a broad range of values, the distance will be governed by this particular feature.

Consider a dataset with two features $\text{age} \in [0, 120]$ and $\text{income} \in [0, 100000]$

Given four points

- $p_1 = (\text{age} = 50, \text{income} = 10000)$
- $p_2 = (\text{age} = 50, \text{income} = 20000), d(p_1, p_2) = 10000.00$
- $p_3 = (\text{age} = 60, \text{income} = 10000), d(p_1, p_3) = 10.00$
- $p_4 = (\text{age} = 60, \text{income} = 20000), d(p_1, p_4) = 10000.00$



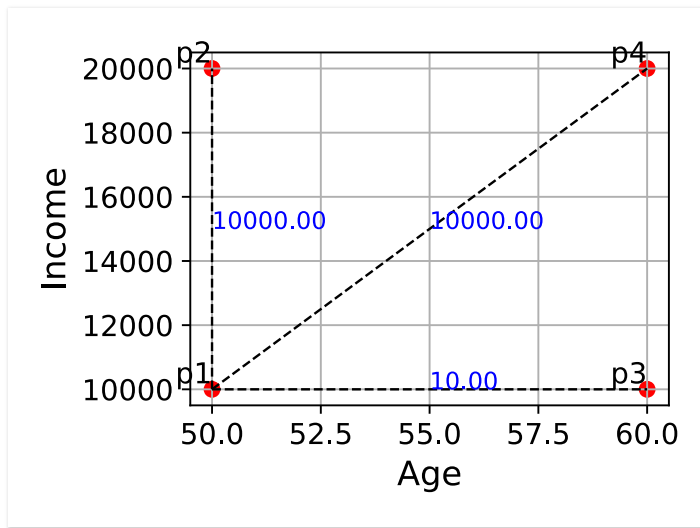
Feature scaling (or data normalization)

Feature scaling normalizes the range of independent variables

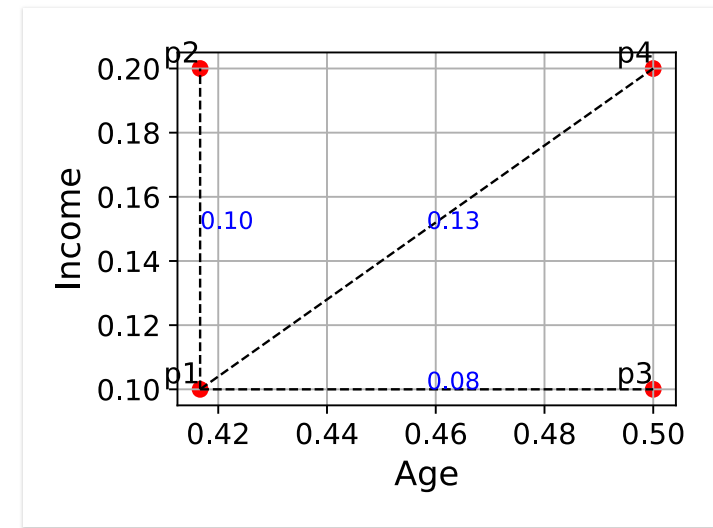
Min-max normalization rescales the features in $[a, b]$ (typically $[0, 1]$): $x' = a + \frac{(x - \min(x))(b - a)}{\max(x) - \min(x)}$

Standardization makes the values of each feature in the data have zero-mean and unit-variance: $x' = \frac{x - \bar{x}}{\sigma}$

Robust scaling is designed to be robust to outliers: $x' = \frac{x - Q_2(x)}{Q_3(x) - Q_1(x)}$

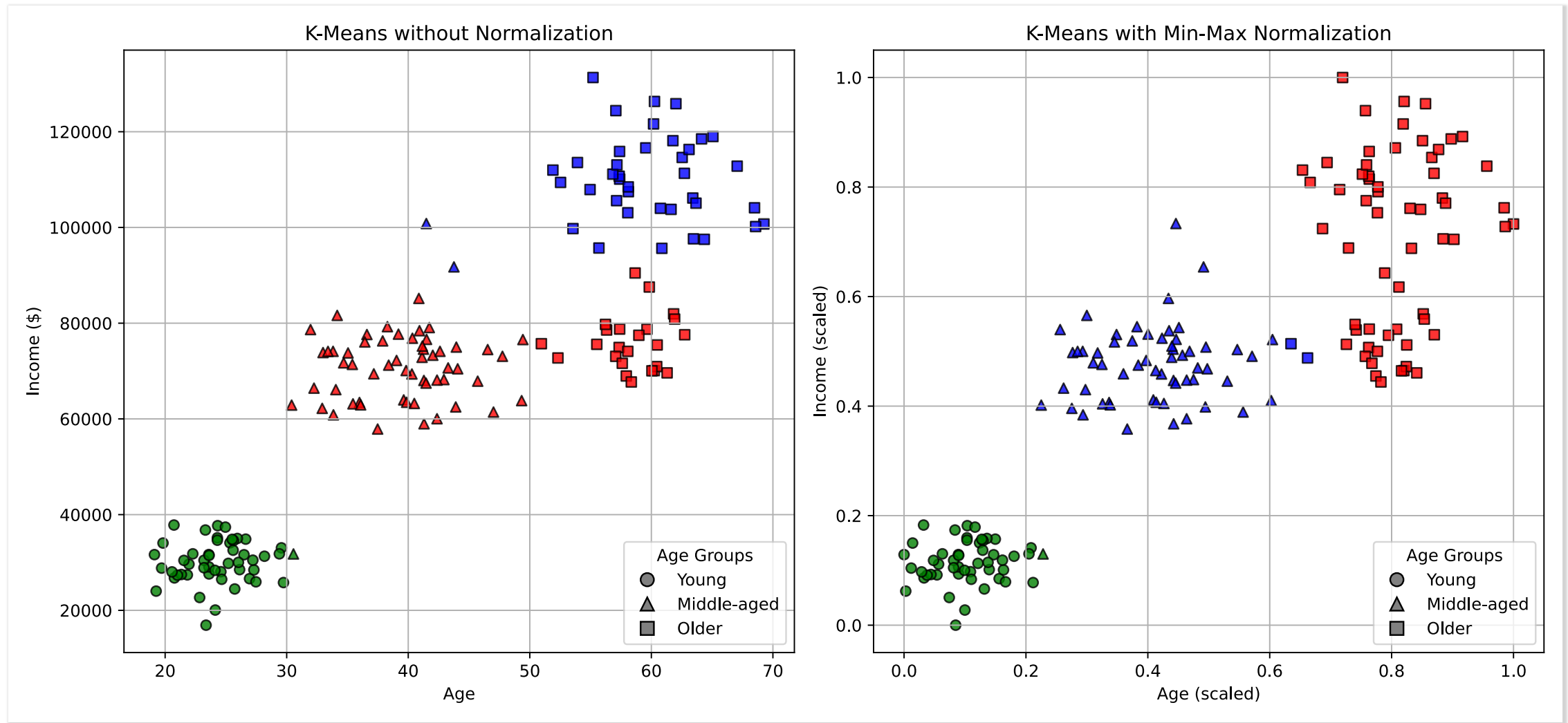


Before min-max normalization



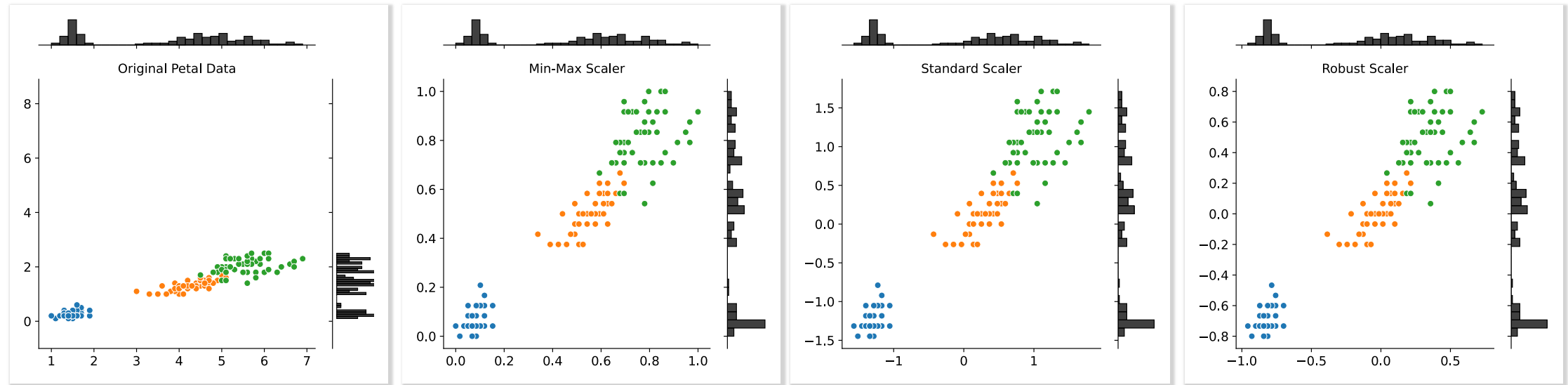
After min-max normalization

Feature scaling

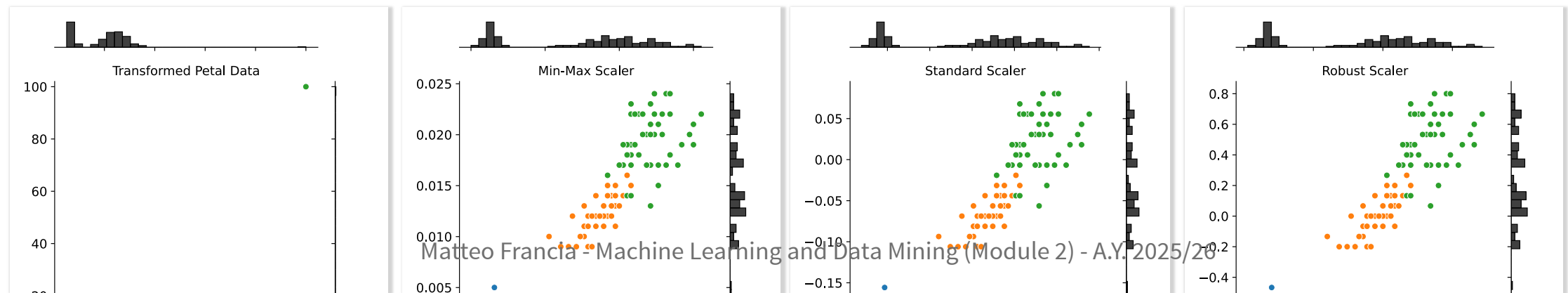


Feature scaling

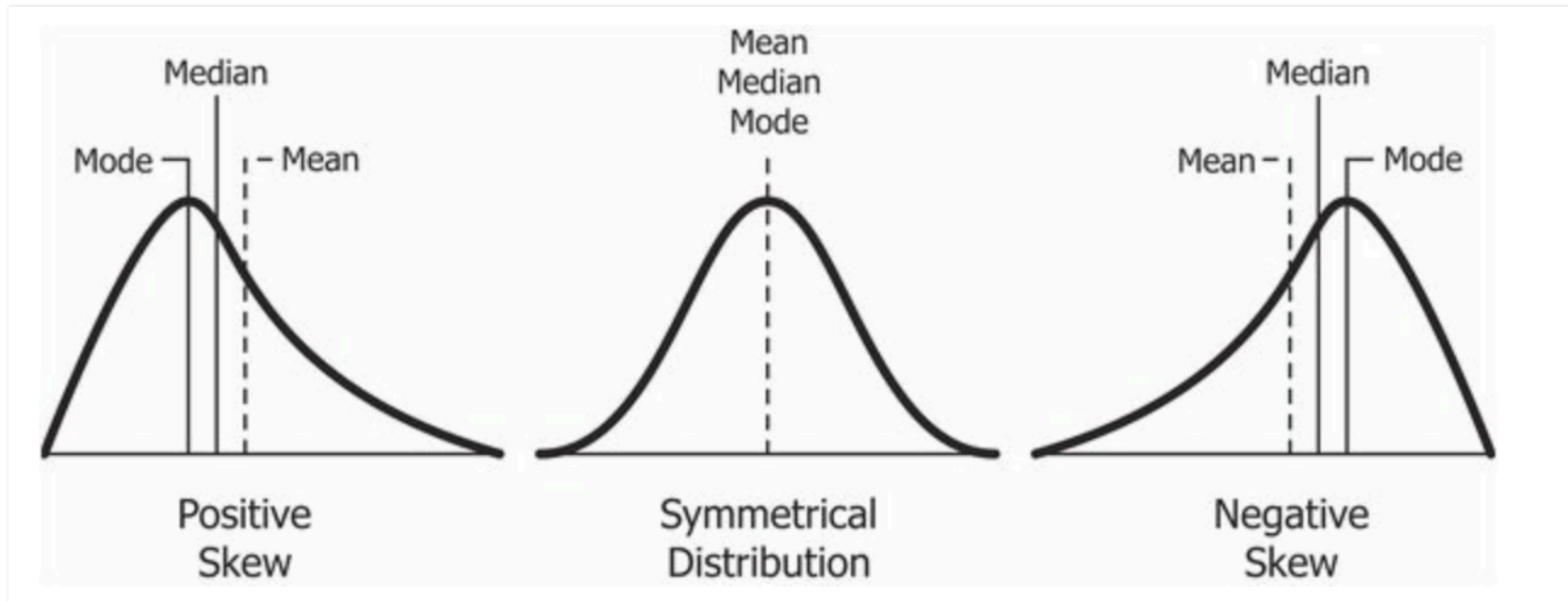
Original Iris dataset



Transformed Iris dataset: `petal_length*=10`, addition of 1 outlier [`petal_length=100`, `petal_width=100`]



What problems can arise with skewed distributions?

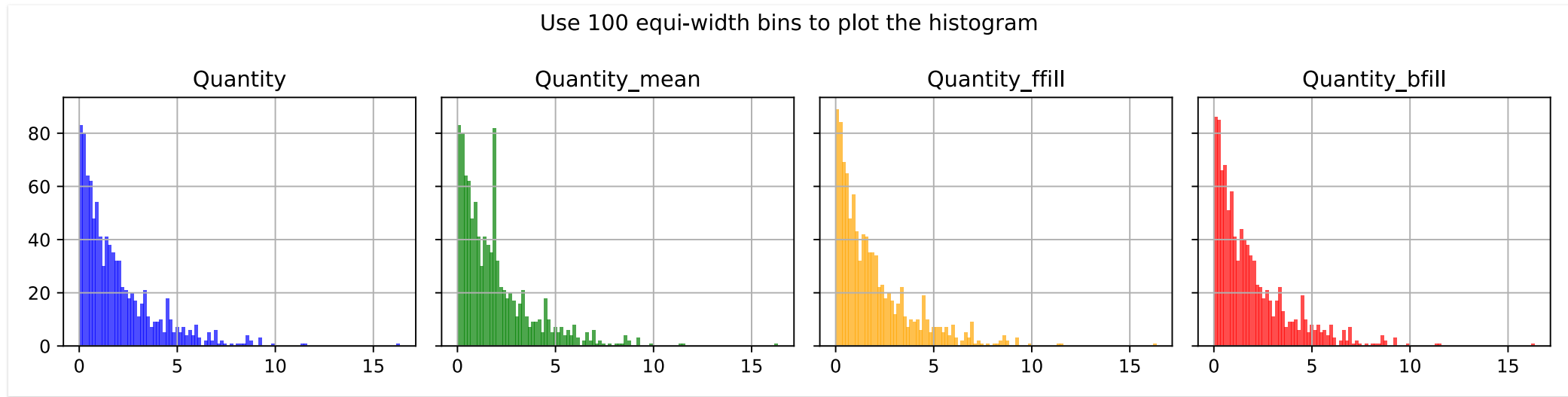


Skewed vs normal distributions

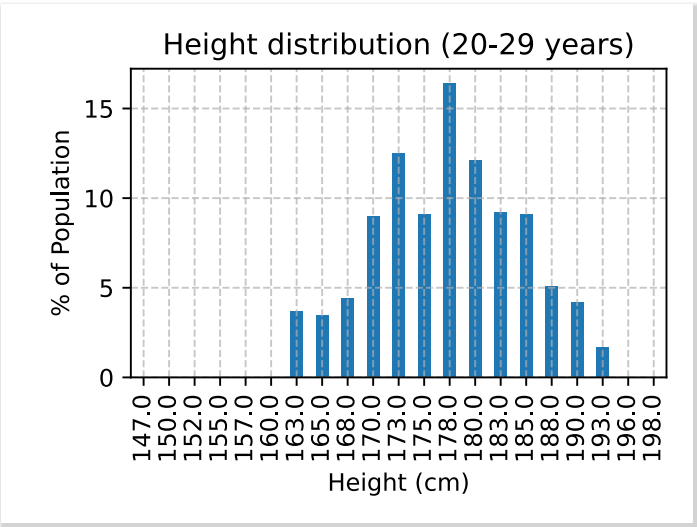
Effects of imputation on skewed distributions?

Long Tail refers to the concept where a large number of niche products collectively generate more sales than a few bestsellers.

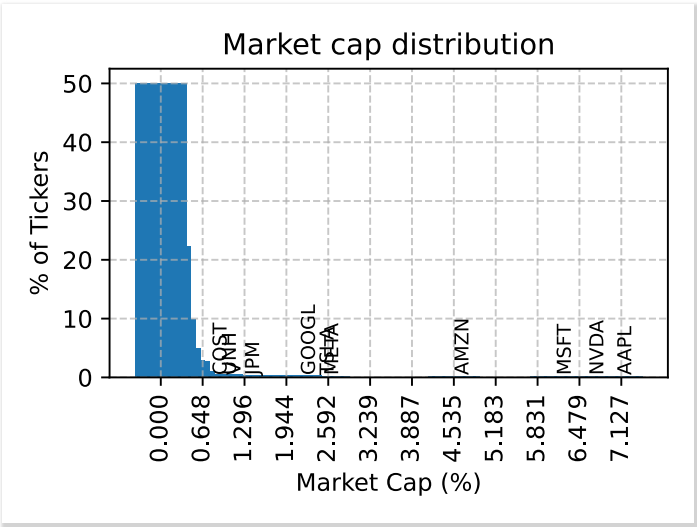
- E-commerce sites such as Amazon stock a vast array of products that traditional retailers wouldn't carry due to space constraints.
- The Long Tail phenomenon is directly related to skewed distributions, specifically a type of right-skewed distribution



Skewed distributions: what happens to mean values?

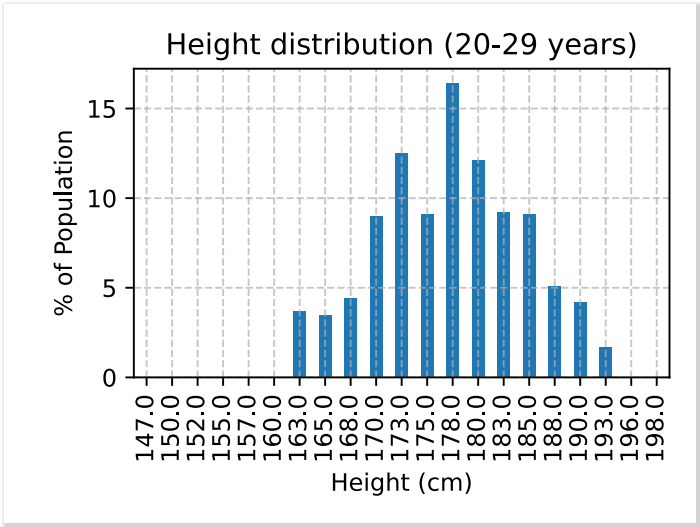


Gaussian distribution



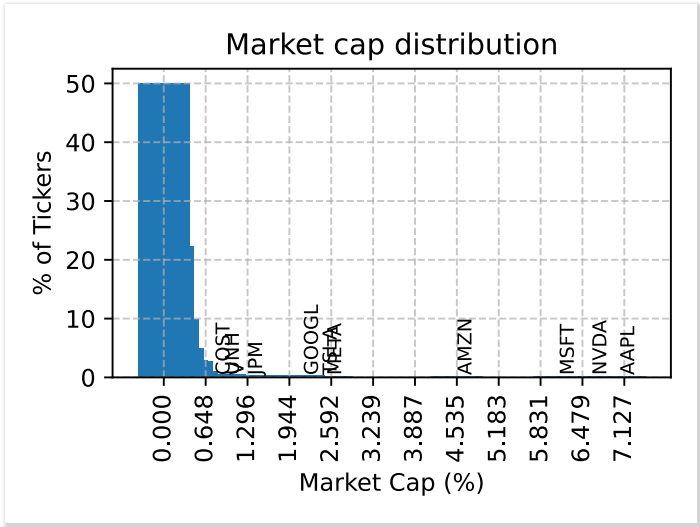
Skewed distribution

Skewed distributions: what happens to mean values?



Gaussian distribution

Mean = 173 cm
Median = 173 cm

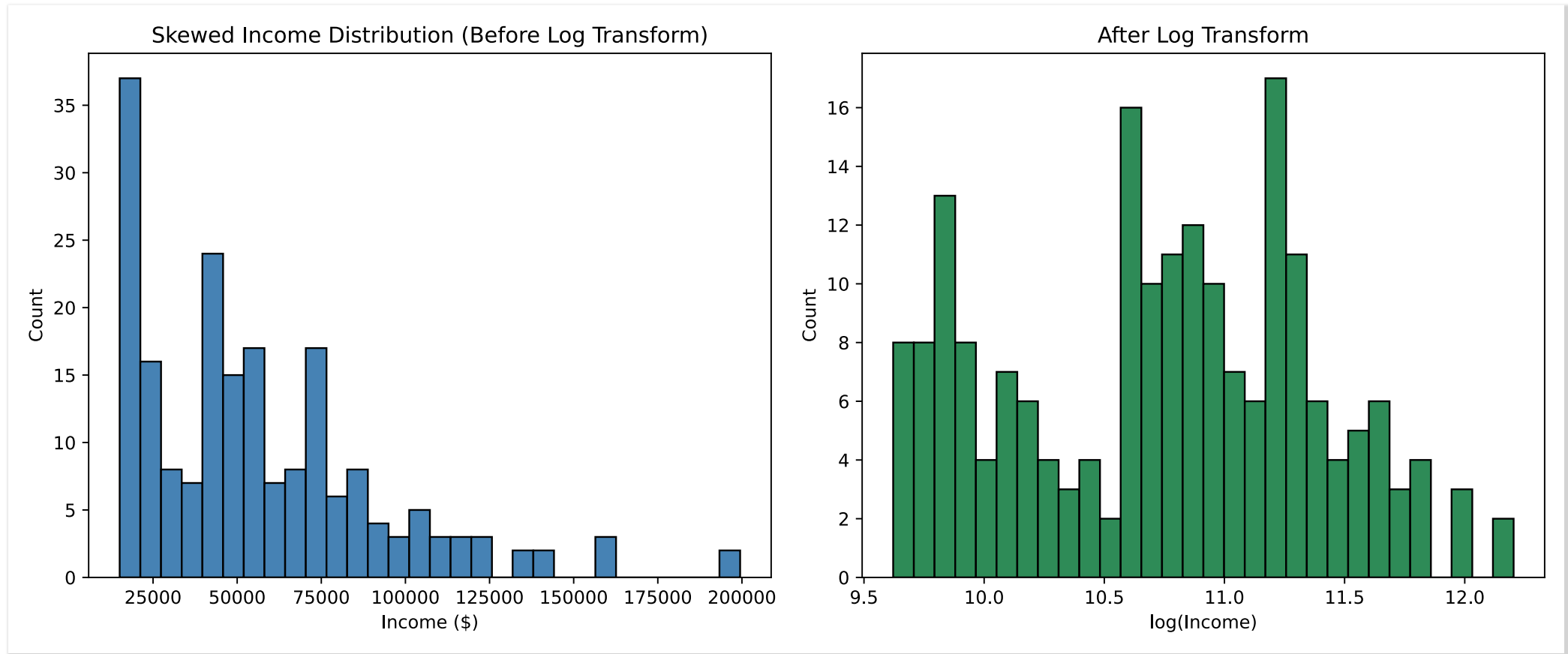


Skewed distribution

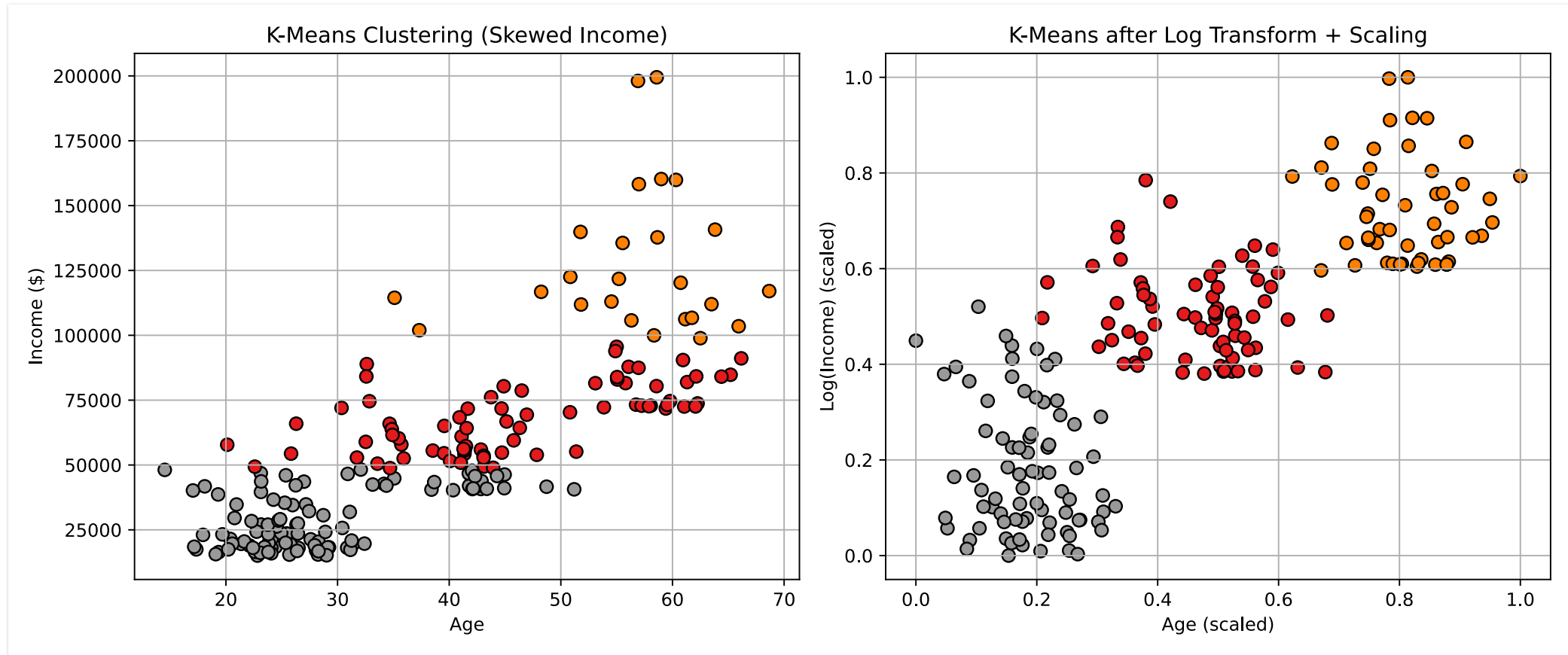
Mean: 103158262914
Median: 36666524821

Skewed distributions

Skewed distributions can be transformed using mathematical functions such as the logarithm.



Skewed distributions





Problem: if the dataset is too detailed/noisy, what can we do?

Some machine learning models can only work with numerical values.

How do we transform the categorical values of the relevant features into numerical ones?

Aggregation and *Binning* may be necessary to *transform ranges to symbolic fields*

Before binning

StoreId	Date	sales
S1	2024-10-04	1000
S1	2024-10-05	1500
S1	2024-10-06	2000

After binning (every 1000€)

StoreId	Date	sales	sales_bin
S1	2024-10-04	1000	[1000-2000)
S1	2024-10-05	1500	[1000-2000)
S1	2024-10-06	2000	[2000-3000)



Problem: if the dataset is too detailed/noisy, what can we do?

A meteorologist analyzes hourly temperature readings, but the data has fluctuations due to temporary weather conditions.

Hour	Temperature (°C)
1	24.1
2	24.3
3	23.8
4	24.5
5	22.9 (Sudden drop due to rain)
6	24.2
7	23.9

How can we smooth small variations?



Problem: if the dataset is too detailed/noisy, what can we do?

For instance, using *equal-width binning* (grouping every 3 hours and averaging):

Time Period	Smoothed Temperature (°C)
1-3 AM	24.0 (Avg of 24.1, 24.3, 23.8)
4-6 AM	23.9 (Avg of 24.5, 22.9, 24.2)
7 AM	23.9

Noise from sudden drops (e.g., 22.9°C at 5 AM) is smoothed, making temperature trends more reliable.

Aggregation

Aggregation computes new values by summarizing information from multiple records and/or tables.

For example, converting a table of product purchases, where there is one record for each purchase, into a new table where there is one record for each store.

Before aggregation (detailed data)

StoreId	ProductId	sales
S1	P1	750
S2	P1	250
S3	P2	...

After aggregation (sum sales by store)

StoreId	sales
S1	1000
S2	1500
S3	...

Pay attention to the *aggregation operator*!

- Correct: sum of sums
 - $(1 + 2) + (3 + 4 + 5) = 1 + 2 + 3 + 4 + 5 = 15$
- Wrong: average of averages
 - $avg(avg(1, 2), avg(3, 4, 5)) = avg(1.5, 4) = 2.75$
 - $avg(1, 2, 3, 4, 5) = 3$

Binning

Data binning is a data pre-processing technique that reduces the effects of minor observation errors

- The original values that fall into a given interval (bin) are replaced by a central value representative of that interval
- Histograms are an example of data binning used in order to observe underlying frequency distributions

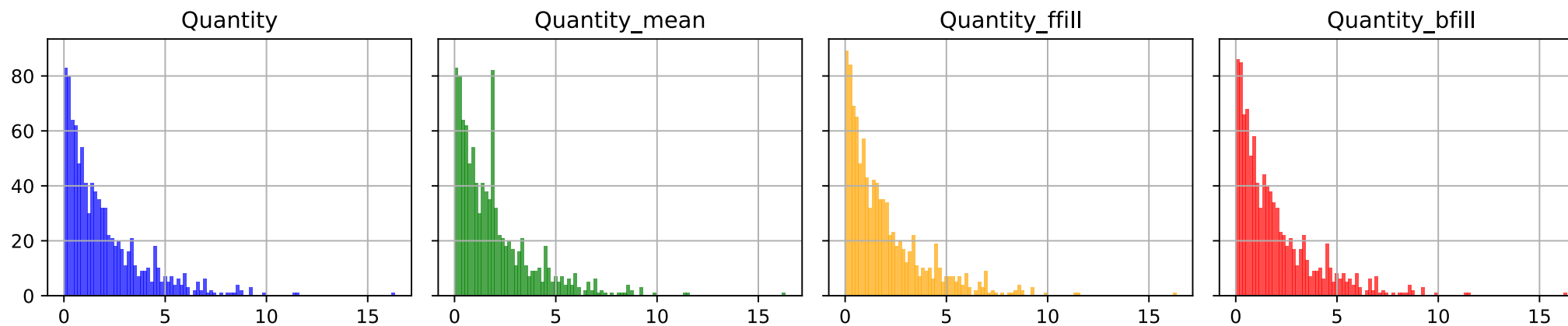
Equal-width: divide the range of values into equal-sized intervals or bins

- For example, if the values range from 0 to 100, and we want 10 bins, each bin will have a width of 10
- It can create empty or sparse bins, especially if the data is skewed or has outliers

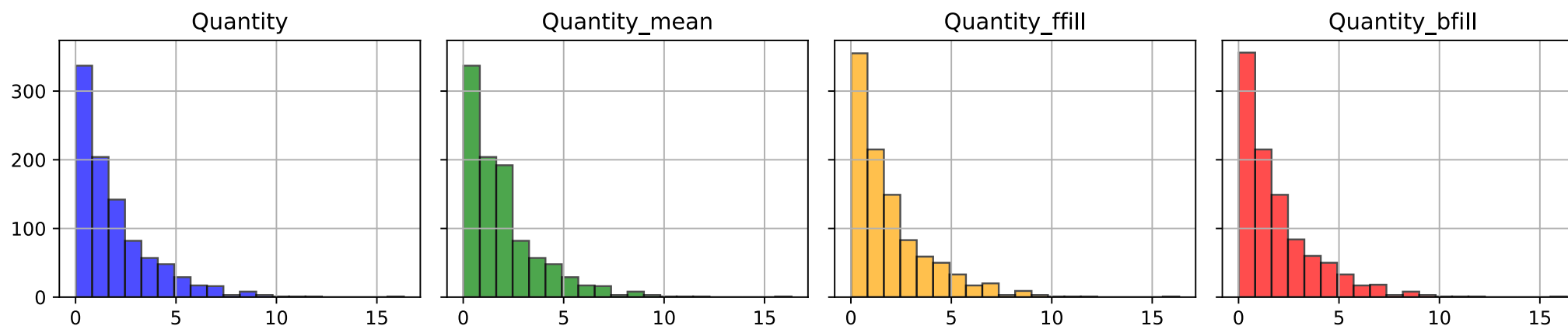
Equal-frequency: divide the values into bins that have the same number of observations or frequency

- For example, if we have 100 observations and we want 10 bins, each bin will have 10 observations
- It creates balanced bins that can handle skewed data and outliers better
- The disadvantage is that it can distort the distribution of the data and create irregular bin widths

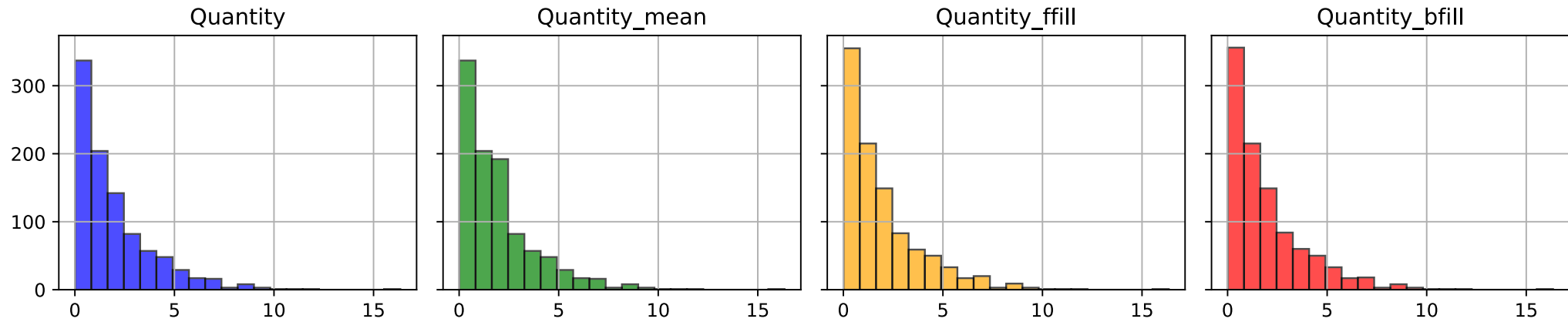
Use 100 equi-width bins to plot the histogram



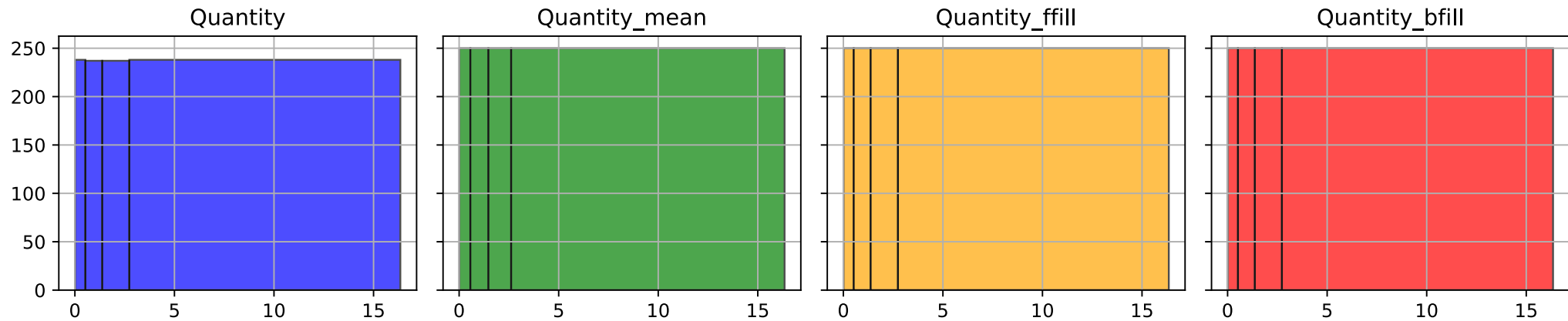
Use 20 equi-width bins to plot the histogram



Use 20 equi-width bins to plot the histogram



Equi-frequency histograms with 4 bins





Problem: what if we have too many features?

A streaming platform wants to recommend movies based on user preferences.

Each movie is represented by a vector of features:

1. `Genre`
2. `Director`
3. `Lead Actor`
4. `IMDB Rating`
5. `Budget`
6. `User Reviews`
7. `Box Office Revenue`
8. `Soundtrack Style`
9. ... and many more (let's assume 100+ features per movie).

If movies had only 2 features (e.g., `Genre` and `IMDB Rating`), we could easily visualize clusters of similar movies.

With 100+ features, the data points are spread out across a vast space.

- All points seem “far apart” from each other, making similarity calculations less reliable.
- Nearest neighbors are not actually close (because all distances become similar).

Dimensionality reduction

Dimensionality reduction is the transformation of data from a high-dimensional space into a low-dimensional space

- Working in high-dimensional spaces can be undesirable for many reasons
- Raw data are often sparse as a consequence of the curse of dimensionality
- Dimensionality reduction can be used for noise reduction, data visualization, cluster analysis, or to facilitate other analyses

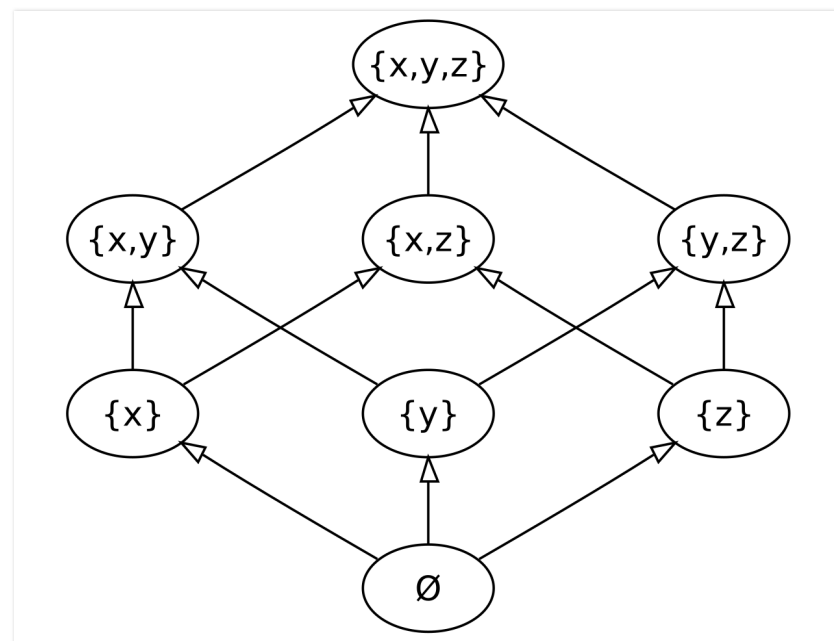
The main approaches can also be divided into *feature selection* and *feature extraction*.

Feature selection

Feature selection is the process of selecting a subset of relevant features (variables, predictors) for use in model construction

Dummy algorithm: test each subset of features to find the one that minimizes the error

- This is an exhaustive search of the space, and is computationally intractable for all but the smallest of feature sets
- If S is a finite set of features with cardinality $|S|$, then the number of all the subsets of S is $|P(S)| = 2^{|S|} - 1$ (do not consider $\emptyset$)
 - With 3 features: $2^3 = 8$ subsets
 - With 4 features: $2^4 = 16$ subsets
 - With 10 features: $2^{10} = 1024$ subsets



Feature selection approaches are characterized by

- *Search technique* for proposing new feature subsets
- *Evaluation measure* for scoring the different feature subsets

Feature selection

Feature selection approaches try to find a subset of the input variables

Filter strategy: select variables regardless of the model

- Based only on general features like the correlation with the variable to predict

Wrapper strategy

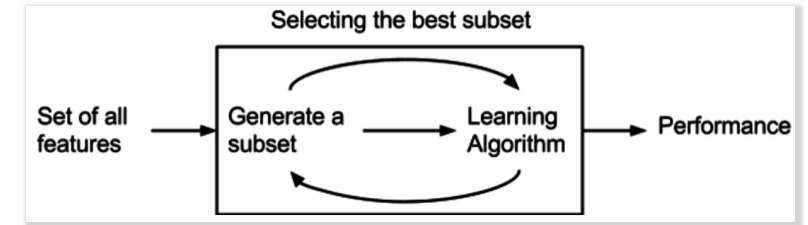
- Methods include forward selection, backward elimination, and exhaustive search

Embedded strategy

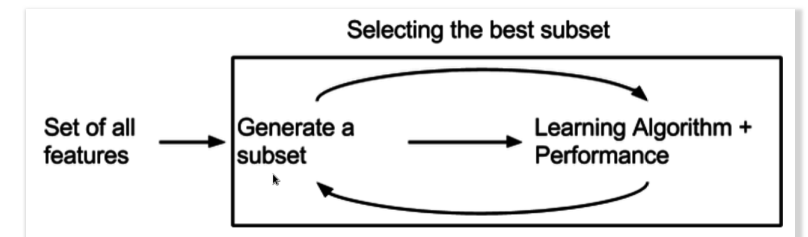
- Add/remove features while building the model based on prediction errors
- A learning algorithm takes advantage of its own variable selection process and performs feature selection and classification simultaneously



Filter



Wrapper



Embedded

Feature selection: Filter strategy

Variance threshold

- Mean: $\mu = \frac{1}{n} \sum_{i=1}^n x_i$, Variance: $Var(X) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$
- Features with low variance do not contribute much information to a model.
- Use a variance threshold to remove any features that have little to no variation in their values.
- Since variance can only be calculated on numeric values, this method only works on quantitative features.

Before selection

StoreId	sales	PostalCode
1	1000	47522
2	1500	47522
3	1000	47522

Compute variance

$VAR(\text{StoreId}) = 0.67$ (?)

$VAR(\text{sales}) = 55555.56$

$VAR(\text{PostalCode}) = 0$

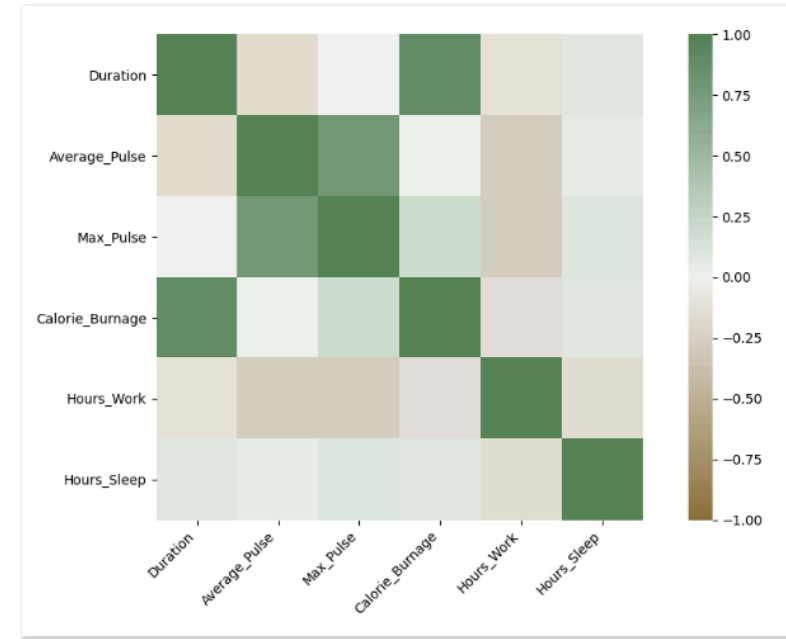
After selection ($VAR(X) > 0.6$)

StoreId	sales
1	1000
2	1500
3	1000

Feature selection: Filter strategy

Pearson's correlation: measures the *linear* relationship between 2 numeric variables

- A coefficient close to 1 represents a positive correlation, -1 a negative correlation, and 0 no correlation
- *Correlation between features*:
 - When two features are highly correlated with one another, then keeping just one to be used in the model will be enough
 - The second variable would only be redundant and serve to contribute unnecessary noise.
- *Correlation between feature and target*:
 - If a feature is not very correlated with the target variable, such as having a coefficient of between -0.3 and 0.3, then it may not be very predictive and can potentially be filtered out.



Feature selection: Wrapper strategy

Each new feature subset is used to train a model, which is tested on a hold-out set

- Counting the number of mistakes made on that hold-out set (the error rate of the model) gives the score for that subset
- As wrapper methods train a new model for each subset, they are very computationally intensive but provide good results

Stepwise regression adds the best feature (or deletes the worst feature) at each round

Backward elimination

- Start with the full model (including all features) and then incrementally remove the most insignificant feature.
- This process repeats again and again until we have the final set of significant features.
 1. Choose a significance level (e.g., $SL = 0.05$ with a 95% confidence).
 2. Fit a full model including all the features.
 3. Consider the feature with the highest p-value.
 - If the p-value $< SL$, terminate the process.
 4. Remove the feature that is under consideration.
 5. Fit a model without this feature. Repeat the entire process from Step 3.

Feature selection: Embedded strategy

Linear regression model: $\hat{y}_i = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$

- Goal is minimizing the sum of squared errors between predicted/actual values
- $\min(\sum_{i=1}^n (y_i - \hat{y}_i)^2)$
 - y_i (red) is the actual value, $\hat{y}_i$ is the predicted value (blue)

Least Absolute Shrinkage and Selection Operator (LASSO)

- Lasso adds a penalty proportional to the absolute values of the coefficients.
- $\min(\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|)$
 - β_j are the coefficients of the model,
 - λ is the regularization parameter controlling the penalty's strength.

Lasso performs automatic feature selection.

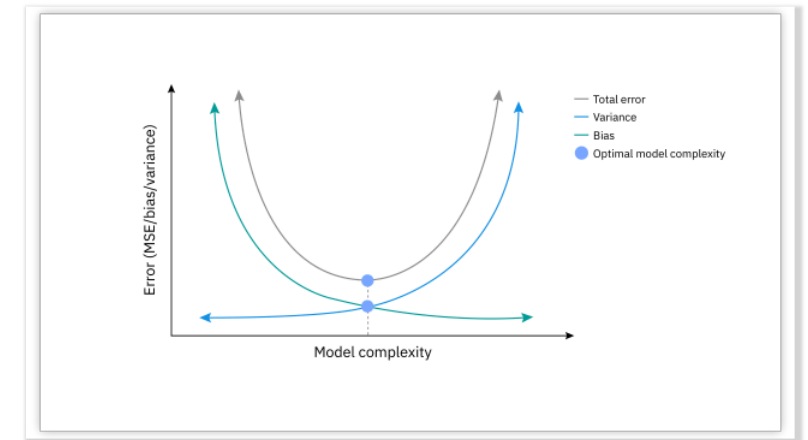
- By shrinking some coefficients to 0, Lasso removes irrelevant features

The optimal λ can be determined with cross-validation techniques.

See also ([Katrutsa and Strijov 2017](#)) ([Chan et al. 2022](#))



Linear regression



Feature extraction (or feature projection)

Feature projection transforms the data from the high-dimensional space to a space of fewer dimensions

- The data transformation may be linear, as in principal component analysis (PCA)
- ... but many nonlinear dimensionality reduction techniques also exist

Principal component analysis (PCA) is a linear dimensionality reduction technique.

- PCA aims to preserve as much of the data's variance as possible in fewer dimensions
- Variance measures how much the data points differ from the mean of the dataset
- The data is linearly transformed onto a new coordinate system such that the directions (*principal components*) capturing the largest variation in the data can be easily identified
- The first principal component captures the highest variance, the second component captures the second highest, and so on.

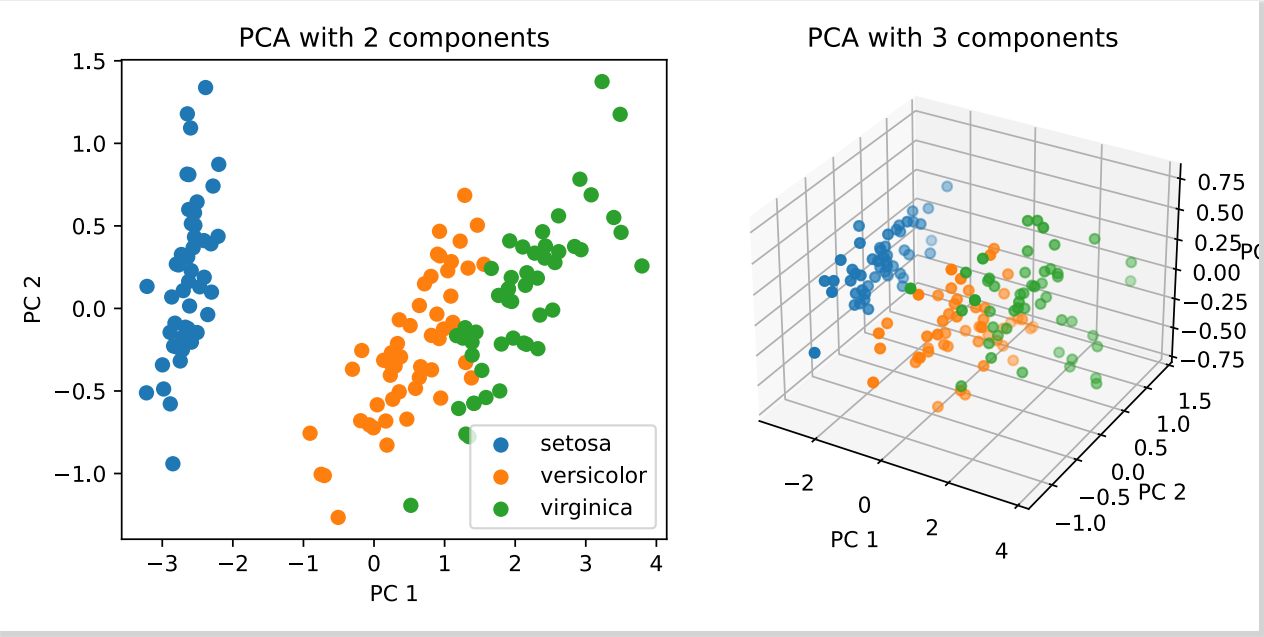
Computing PCA

- PCA is sensitive to the scale of the data. The first step is usually to standardize the features (mean = 0, standard deviation = 1) to ensure that all features contribute equally to the analysis.
- Then, compute the covariance Matrix
 - Eigenvectors represent the directions of the principal components.
 - Eigenvalues represent the magnitude of variance in the direction of the corresponding eigenvector.
- The eigenvector with the largest eigenvalue is the first principal component, and so on.

PCA on the Iris dataset

Iris contains 4 features; we cannot plot it directly.

- 1. petal_length
- 2. petal_width
- 3. sepal_length
- 4. sepal_width



Principal Component	Explained Variance
PC 1	92.46%
PC 2	5.31%
PC 3	1.71%

Feature Relevance for 3 Components:

Feature	PC 1	PC 2	PC 3
Sepal Length (cm)	0.361	0.657	-0.582
Sepal Width (cm)	-0.085	0.730	0.598
Petal Length (cm)	0.857	-0.173	0.076
Petal Width (cm)	0.358	-0.075	0.546



Problem: how do we integrate different data sources?

A hospital wants to analyze patient health records by integrating data from multiple sources, including electronic health records (EHRs), wearable devices, and insurance claims.

Integrate Data

Integration involves *combining information from multiple tables* or records to create new records or values.

- With table-based data, an analyst can join two or more tables that have different information about the same objects.

For instance, a retail chain has one table with information about each store’s general characteristics (e.g., floor space, type of mall), another table with summarized sales data (e.g., profit, percent change in sales from the previous year), and another table with information about the demographics of the surrounding area.

These tables can be merged together into a new table with one record for each store.

StoreId	Type
S1	grocery
S2	supermarket
S3	...

+

StoreId	Sales
S1	1000
S2	1500
S3	...

=

StoreId	Type	Sales
S1	grocery	1000
S2	supermarket	1500
S3	...	...

Data integration

Data integration combines data residing in different sources and provides users with a unified view of them.

Primary key-based integration combines multiple sources based on matching unique identifiers (primary keys).

- This method works when both datasets have a well-defined and consistent schema with common key fields.

Semantic integration focuses on understanding the meaning of the data from different sources to combine it effectively.

- The goal is to merge data that may use different names, terminologies, or structures to describe the same concepts.
- Data is integrated based on semantic meaning rather than structural similarities.
- It involves the use of ontologies or data dictionaries to map similar concepts across datasets, ensuring consistency.
- It requires understanding the context, meaning, and relationships within the data.
 - For instance, spatial data can be easily integrated into maps

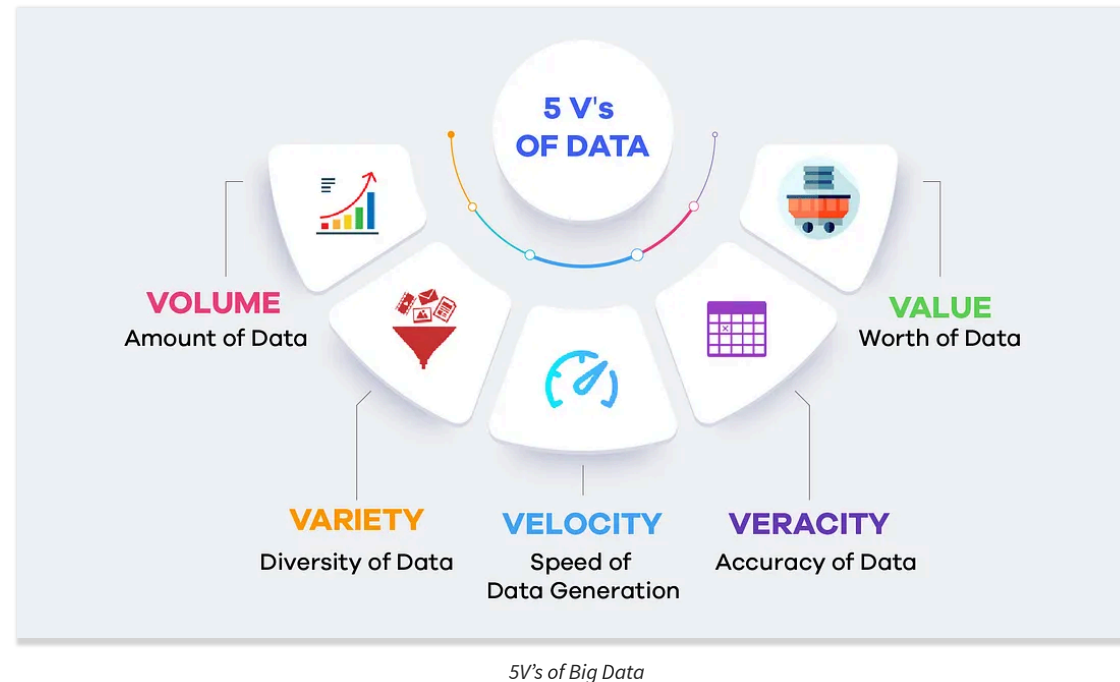
Semantic Integration vs Primary Key-based Integration

Aspect	Semantic Integration	Primary Key-based Integration
<i>Approach</i>	Based on meaning and understanding of the data.	Based on matching unique keys.
<i>Suitability</i>	Data with heterogeneous terminologies or structures.	Datasets have common, well-defined keys.
<i>Complexity</i>	Complex to interpret and align meanings.	Simpler, relies on exact key matches.
<i>Flexibility</i>	Integrate data with different schemas/representations.	Less flexible, requires shared primary key fields.
<i>Challenges</i>	Requires mapping of concepts and domain semantics.	Limited to datasets that share a key.

Format Data

In some cases, the data analyst will *change the format (structure) of the data*.

- Sometimes these changes are needed to make the data suitable for a specific modeling tool.
- In other instances, the changes are needed to pose the necessary data mining questions.



Examples:

- Simple: removing illegal characters from strings or trimming them to a maximum length
- More complex: reorganization of the information (e.g., *from normalized to flat tables*)

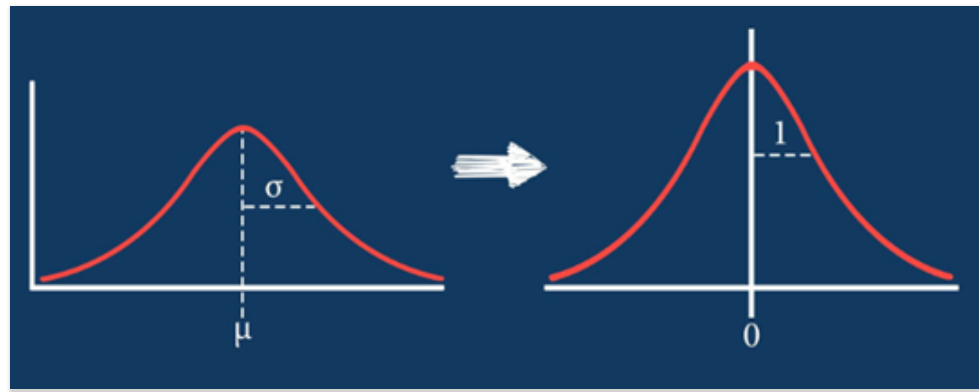


Problem: how do we concatenate pre-processing transformations?

Sequences of transformations

Things are even more complex when applying sequences of transformations.

- E.g., normalization should be applied before rebalancing since rebalancing can alter average and standard deviations
- E.g., applying feature engineering before/after rebalancing produces different results depending on the dataset and algorithm



image

More an art than a science

- ... At least for now

Final considerations

Overlapping with Business Intelligence and Data Warehousing

ETL (Extract, Transform, Load) is one of the most widely used data integration techniques in data warehousing.

- *Extract*: Pull data from multiple sources (e.g., databases, APIs, flat files).
- *Transform*: Clean, standardize, and transform the data into the desired format.
- *Load*: Load the transformed data into a target database or data warehouse.

ELT (Extract, Load, Transform) loads data into a storage system (like a data lake) and then transforms within the storage system.

Overlapping with Big Data and Cloud Platforms

- Data *profiling* to get metadata summarizing our dataset
- Data *provenance* to track all the transformations that we apply to our dataset

Wooclap

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EN ▼

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